

AMENDMENT NO. 1 FEBRUARY 1988
TO
IS : 4129 - 1967 METHODS OF CHEMICAL
ANALYSIS OF ALUMINIUM TRIFLUORIDE

(Page 13, clause 8.4.2) — Add the following new clauses after 8.4.2:

9. DETERMINATION OF LOSS ON IGNITION

9.1 Reagents

9.1.1 *Magnesium Perchlorate (Anhydrous)*

9.2 Procedure

9.2.1 Weigh 1.00 g of sample into a previously weighed platinum crucible. Heat slowly first and ignite in a furnace at a temperature 550°C for half an hour. Take it out, cool in a desiccator containing preferably magnesium perchlorate and weigh to constant weight.

9.3 Calculation

$$\text{Loss on ignition} = \frac{A - B}{C} \times 100$$

where

A = mass of crucible with sample,

B = mass of crucible with residue after ignition, and

C = mass of sample taken.

10. DETERMINATION OF SULPHATE (AS SO₃) BY GRAVIMETRIC METHOD

10.1 Outline of the Method — The sample is fused with sodium carbonate and boric acid. The sulphate is precipitated by barium chloride, ignited and weighed.

10.2 Reagents

10.2.1 *Sodium Carbonate Anhydrous*

10.2.2 *Boric Acid*

10.2.3 *Concentrated Hydrochloric Acid* — (*rd* = 1.16) (Conforming to IS : 265-1976*.)

*Specification for hydrochloric acid (*second revision*).

10.2.4 Dilute Ammonia Solution — 1 : 1 (v/v).

10.2.5 Methyl Red Indicator Solution — 0.1 percent in methanol.

10.3 Procedure

10.3.1 Weigh 1.00 g of sample and mix with 4 g of sodium carbonate and 2 g of boric acid in a platinum crucible. Fuse over a burner or in a muffle furnace heated to 1 000°C for 15-20 minutes.

10.3.2 Cool, place the crucible in a 500-ml beaker, add 100 ml water and 20 ml concentrated hydrochloric acid and heat to dissolve the melt. Wash the crucible with hot water and remove from the beaker.

10.3.3 Add 50 ml water and some filter paper pulp and filter through Whatman No. 41 filter paper and wash with hot water.

10.3.4 Dilute the filtrate and washings to 250 ml, heat to boil. Neutralise the excess acid with dilute ammonium hydroxide solution using methyl red indicator. Acidify the solution again by adding 2-3 ml of concentrated hydrochloric acid. Add slowly 10 ml of barium chloride solution and stir. Allow to stand for overnight. Filter through Whatman No. 42 filter paper and wash well with the hot water till free from chloride. Ignite the precipitate in a weighed platinum crucible at 600°C and then finally at 800°C. Cool in a desiccator and weigh the residue to the content weight.

10.3.5 Blank — Carry out a reagent blank using the procedure given above without the sample.

10.4 Calculation — Calculate the percentage of sulphate as follows:

$$\text{Sulphate (as SO}_3 \text{) , percent by mass} = \frac{(A - B)}{C} \times 34.30$$

where

A = mass in g of barium sulphate obtained in the sample,

B = mass in g of barium sulphate obtained in the blank, and

C = mass in g of sample taken.

11. DETERMINATION OF PHOSPHOROUS

11.1 Outline of the Method — Sample is fused with sodium carbonate and boric acid and fused mass is treated with water. The sample solution is adjusted to pH 1.6 to 2.0 and treated with ammonium molybdate to form a complex and reduced with stannous chloride and measured at 710 nm in a spectrophotometer.

11.2 Reagents

11.2.1 Sodium Carbonate, Anhydrous

11.2.2 Boric Acid

11.2.3 Dilute Sulphuric Acid — 1 : 3 (v/v).

11.2.4 Hydroxyl Amine Hydrochloride — 2 percent.

11.2.5 Ammonium Molybdate Solution — Dissolve 3 g of ammonium molybdate in 60 ml concentrated hydrochloric acid and make up to 100 ml with distilled water.

11.2.6 Stannous Chloride (Stock Solution) — Dissolve 10 g of stannous chloride in 25 ml of concentrated hydrochloric acid.

11.2.6.1 Stannous chloride (working solution) — Dilute 1.5 ml of stock solution to 250 ml with distilled water.

11.2.7 Standard Phosphate Solution — Weigh 0.1917 g of dried potassium dihydrogen phosphate (KH_2PO_4), transfer to a 1000-ml volumetric flask and make up to the mark (1 ml = 0.1 mg P_2O_5).

11.2.7.1 Dilute 10 ml of the above solution to 100 ml in a volumetric flask (1 ml = 0.01 mg P_2O_5).

11.3 Procedure

11.3.1 Weigh 1 g of sample, mix with 2 g of boric acid and 4 g of sodium carbonate in a platinum crucible. Incinerate the contents from low bunsen flame to muffle furnace. Swirl the crucible and contents and heat again for 10 minutes. Remove the crucible from the furnace and cool. Fill the crucible with water. Heat gently until the entire melt has dissolved, then cool the solution and transfer to a 250-ml beaker containing about 100 ml of water. Rinse the crucible with 10-20 drops of dilute sulphuric acid and transfer to the beaker with few millilitres of water. Adjust to pH 1.6 to 2.0 with dilute sulphuric acid.

11.3.1.1 Add 5 ml of hydroxylamine hydrochloride solution and add 10 ml of ammonium molybdate solution. After 10 minutes add 5 ml of stannous chloride solution (working solution). Transfer the contents to 250-ml volumetric flask. After 10 minutes measure the absorbance of the solution against a reagent blank in a spectrophotometer at 710 nm.

11.3.2 Blank — Weigh and mix 2 g of boric acid and 4 g of sodium carbonate. As described in the sample fusion, dissolve the melt, transfer to a 250-ml beaker containing about 100 ml of water and adjust pH 1.6 to 2.0. Proceed in accordance with **11.3.1.1**.

11.3.3 Calibration Curve — Transfer 0, 2, 4, 6, 10, 15 and 20 ml of phosphate solution in seven 250-ml beakers. Proceed in accordance with **11.3.1.1** and plot the photometric readings of calibration solution against mg of P_2O_5 /250 ml solution.

11.3.4 Calculation — Convert the photometric readings of sample and blank solution to mg of phosphate by means of calibration curve and calculate the percentage of phosphorous (as P_2O_5) as follows:

$$\text{Phosphorus (as } P_2O_5 \text{)} = \frac{A - B}{C} \times \frac{1}{10}$$

where

A = mass in mg of P_2O_5 found in sample,

B = mass in mg of P_2O_5 found in blank, and

C = mass in g of sample taken for the test.

(SMDC 35)